

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

2nd Semester of M. Pharm (Pharmaceutical Technology)

University Theory Examination April May 2019

PHTCH005 Molecular Pharmaceutics

Date: 25-04-2019, Thursday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 75

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1 (A) Write a note on pMDI. 05

(B) What is drug targeting? What are the levels of drug targeting? 05
Differentiate between active & passive drug targeting.

Q.2 Answer the followings: (Any three) 15

(A) Describe the factors affecting drug transport across the blood brain barrier.

(B) What are monoclonal antibodies? Describe the method for the production of monoclonal antibodies.

(C) What are the applications, advantages and disadvantages of niosomes?

(D) What are aptamers? Describe its classification and applications.

Q.3 Answer the followings: (Any three) 15

(A) Physical methods of gene delivery

(B) Evaluation of dry powder inhaler

(C) Applications of monoclonal antibodies

(D) Characterization of cyclosporine microemulsion

SECTION-II

Q.4 Define preformulation. What are the objectives of preformulation study? **10**
Discuss in detail preformulation study to be carried out for development of oral solid dosage form.

Q-5 What are liposomes? How they are classified? Enumerate the methods for preparation of liposomes and explain any one method in detail. **10**

OR

Q-5 (A) Write note on applications of intranasal drug delivery. **05**

(B) Enumerate the methods for the preparation of microspheres. Describe any one method in details. **05**

Q-6 Answer the followings: (Any three) **15**

(A) What are phytosomes? How they are prepared? What are its applications?

(B) What are antisense oligonucleotides? What are its types? Describe its applications.

(C) What are polymeric nanoparticles? Describe the objectives for preparation of nanoparticles and its applications.

(D) Write note on factors affecting nasal drug absorption.